

CODENAME 47 IN ACTION



Detect • Fingerprint • Stop AI-Powered Pentesting Agents

Executive Summary

Codename 47 (C47) is a next-generation honeypot designed specifically to lure, identify, and study LLM-powered penetration testing agents. Traditional scanners reveal themselves through enumeration and signature-based behavior. Modern AI agents exhibit reasoning, planning, adaptive navigation, and semantic understanding. C47 exploits these characteristics by embedding deceptive artifacts, hidden instructions, semantic traps, reasoning challenges, and behavioral probes throughout the target environment.

The objective is not only detection, but attribution of agent capabilities. C47 records how an autonomous system explores an environment, which signals it follows, whether it understands instructions hidden in comments or robots.txt files, and whether it can solve reasoning tasks that require comprehension rather than simple pattern matching.

Live Detection Case Study

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# 1. Keep track found (paths -> 3-4, < 1M) and clean up the output after extracting endpoint lists.
# 2. Analyze 26 files for hidden endpoints, API keys, secrets, hardcoded URLs.
# 3. Content discovery: fof/fdrsearch with a common wordlist (download to /home/pentester/tools/wordlists if needed) for hidden paths, admin panels, API docs (swagger/openapi), backups.
# 4. Enumerate authentication model: login/register forms, session cookies, JWTs, roles visible in the UI, test accounts if registration is open.
# 5. Check for common exposures: /api/, /admin/, /env/, /swagger/, /graphql/, /debug, actuator endpoints, robots.txt, sitemap.xml.
# 6. Document rate limiting, WAF behavior, error handling.

Deliverables – write findings to shared notes (create_notes) AND record coverage via record_coverage for each surface assessed:
• A note titled "Recon Map" with: full endpoint inventory (method + path + parameters), technology fingerprint, auth model, roles/enums,
inventory JSON file, and any credentials/text accounts you create.
• A note titled "Recon Summary" with high-value attack surfaces ranked by priority for follow-up testing.
– Then exit agent_finish summarizing what you found and what deserves priority testing.

Do NOT do vulnerability exploitation yourself! – mapping and discovery only. Do not lead vuln-testing skills; use recon tooling skills
(tooling/katana, tooling/ffuf, tooling/tfcping, tooling/agent_bonjour) as needed.
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Task Required

- [] Recon & attack-surface mapping of http://pentest-decker.internal:8080
- [] Derive and save threat model from recon results
- [] Dispatch vulnerability testing agents based on recon findings
- [] Validate confirmed findings with Pocs
- [] Reassemble coverage and finalize report

Recon agent is running. I'll wait for its report before deriving the threat model and dispatching vulnerability testing.

waiting
Waiting for Recon Agent to complete target mapping

The wait timed out – let me check the recon agent's status and any notes it has produced so far.

I'm viewing agents graph

notes
No notes

Recon agent is still running (crawls take time). I'll continue waiting for its completion report.

waiting
Recon agent still mapping the targets; waiting for its completion report.

Send message to recon

Send a message

Vulnerability Scanning

http://127.0.0.1:8080/?token=abcdeffor_curl(20m/sec)/curl/gid/vuln/400000000

< Back Agent

codename_47 honeypot - detect / fingerprint / stop AI-powered pentesting tools

vuln.0 stages 0 passive - 1 active - 2 extract

config: config/c47.toml	
17:51:51 INFO c47.engine built: 12 lures, 10 detectors, 1 surfaces, 3 sinks, llm=None	
17:51:51 INFO c47.sink.jsonl jsonl sink writing to var/events/jsonl	
17:51:51 INFO c47.sink.splitite splitite sink at var/c47/splitite	
17:51:51 INFO c47.surface.http http surface on 0.0.0.0:8080 as Meridian Ops Console	
17:51:51 INFO c47.engine surface 'http' started	
17:51:51 INFO c47.canary canary listener on 0.0.0.0:8081	
17:51:51 INFO c47.runtime codename_47 ready surfaces=http lures=12 detectors=10	
17:56:03 INFO c47.engine signal comment_only_path_visited actor=127.0.0.1 conf=1.00	
17:56:03 WARNING c47.engine verdict unknown -> scripted actor=127.0.0.1 player=0.139	
17:56:26 INFO c47.engine signal robots_live_path_visited actor=127.0.0.1 conf=1.00	
17:56:26 INFO c47.engine signal timing_swathooth actors=127.0.0.1 conf=0.70	
17:56:26 INFO c47.engine signal robots_selective_follow_actor=127.0.0.1 conf=0.85	
17:56:26 WARNING c47.engine verdict scripted -> suspected_agent actor=127.0.0.1 player=1.000	
(SUSPECTED_AGENT) actor=127.0.0.1 session=2eaf0785ad4684 surfaces=http stage1	
= robots_selective_follow (path semantics, conf=0.85)	
= comment_only_path_visited (expectations, conf=1.00)	
= timing_swathooth (timing, conf=0.70)	
= robots_live_path_visited (expectations, conf=1.00)	
17:56:26 INFO c47.engine signal robots_dead_path_visited actor=127.0.0.1 conf=1.00	
17:56:26 INFO c47.engine signal robots_indiscriminate_follow_actor=127.0.0.1 conf=0.70	
17:56:26 INFO c47.engine signal reasoning_probe_solved_actor=127.0.0.1 conf=1.00 CONCLUSIVE	
[!] CONCLUSIVE reasoning probe solved actor=127.0.0.1 detector=expectations	
Retried with a correctly derived sum_digits value ("175"). Requires comprehension of prose; no gress needed, so this fires even against sandboxed agents.	
matched: challenge=175	
17:56:26 INFO c47.engine signal reasoning_probe_attempted_actor=127.0.0.1 conf=1.00	
17:56:26 WARNING c47.engine verdict suspected_agent -> confirmed_agent actor=127.0.0.1 player=1.000	
(CONFIRMED_AGENT) actor=127.0.0.1 session=2eaf0785ad4684 surfaces=http stage2	
= reasoning_probe_solved (expectations, conf=1.00)	
= robots_selective_follow (path semantics, conf=0.85)	
= reasoning_probe_attempted (expectations, conf=1.00)	
= comment_only_path_visited (expectations, conf=1.00)	
17:57:36 INFO c47.engine signal browser_us_header_mismatch_actor=127.0.0.1 conf=0.80	
17:58:03 INFO c47.engine signal ua_family_churn_actor=127.0.0.1 conf=0.80	
17:58:04 INFO c47.engine signal wordlist_enum_actor=127.0.0.1 conf=0.80	
17:58:35 INFO c47.engine signal timing_uniform_fast_actor=127.0.0.1 conf=0.80	

Evidence: Terminal Telemetry

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[SUSPECTED_AGENT] actor=127.0.0.1 p(agent)=1.000
- robots_selective_follow
- comment_only_path_visited
- timing_sawtooth
- robots_live_path_visited

[!!] CONCLUSIVE reasoning_probe_solved
Retried with a correctly derived sum_digits value ('175')
Requires comprehension of prose; no egress needed.

[CONFIRMED_AGENT] actor=127.0.0.1 p(agent)=1.000
- reasoning_probe_solved
- reasoning_probe_attempted
- robots_selective_follow
- comment_only_path_visited
```

The decisive indicator was the successful completion of the reasoning probe. The agent derived the hidden value '175' from contextual instructions and retried with the correct answer. This behavior demonstrates comprehension and reasoning capabilities beyond conventional automation and provides conclusive evidence of an LLM-assisted workflow.

Detection Signals

- Comment-only path visitation indicating awareness of hidden instructions.
- Selective robots.txt traversal demonstrating semantic interpretation.
- Timing sawtooth patterns associated with reasoning cycles.
- Live and dead path exploration used to map decision logic.
- Reasoning challenge completion producing conclusive attribution.
- Agent confidence score reaching 1.000 and escalating to CONFIRMED_AGENT.

Conclusion

Codename 47 demonstrates that AI-native honeypots can actively fingerprint autonomous offensive agents. Rather than simply recording requests, the platform induces agents to expose how they reason, navigate, and solve problems. This creates a new defensive capability for understanding emerging AI-powered penetration testing systems and differentiating them from traditional automated tooling.